Magistein notes:

SC: $$H=\sum_k (c_k \;\; c^\dagger_{-k})\underbrace{\begin{pmatrix} -\xi_k & -\Delta^*_k \\ -\Delta_k & \xi_k\end{pmatrix}}_{\mathcal{H}_k}\begin{pmatrix} c^\dagger_k \\ c_{-k}\end{pmatrix}\underbrace{\left(-\sum_{k k'} V_{kk'}\langle c^\dagger_k c^\dagger_{-k}\rangle\langle c_{-k'} c_{k'}\rangle + \sum_k \xi_k\right)}_{k_0 \text{ - energy shift}}$$

$$\Delta_k = \sum_{k'} V_{kk'}\langle c_{-k'} c_{k'}\rangle \;, \quad \Delta^*_k = \sum_{k'} V^*_{kk'}\langle c^\dagger_{k'} c^\dagger_{-k'}\rangle$$

diagonalizacija:

① Bogoljubov:

$$\alpha_k = u_k c_k - v_k c^\dagger_{-k} \qquad \alpha_{-k} = u_k c_{-k} + v_k c^\dagger_k$$
$$\alpha^\dagger_k = u_k c^\dagger_k - v_k c_{-k} \qquad \alpha^\dagger_{-k} = u_k c^\dagger_{-k} + v_k c_k$$

| Bogoljubov

↓ inverse

$$c_k = u_k\alpha_k + v_k\alpha^\dagger_{-k} \qquad c_{-k} = u_k\alpha_{-k} - v_k\alpha^\dagger_k$$
$$c^\dagger_k = u_k\alpha^\dagger_k + v_k\alpha_{-k} \qquad c^\dagger_{-k} = u_k\alpha^\dagger_{-k} - v_k\alpha_k$$

• $u_k, v_k \in \mathbb{R}$, $u_k = u_{-k}$, $v_k = v_{-k}$ → CCR: $\boxed{u_k^2 + v_k^2 = 1}$

je že $K = H - \mu N$!

$$H = \sum_k \xi_k (c^\dagger_k c_k + c^\dagger_{-k} c_{-k}) - \bar V \sum_{kk'}{}' \underbrace{c^\dagger_{-k'} c^\dagger_{k'} c_k c_{-k}}_{\substack{\| \\ b^\dagger_k b_{k'}}}$$

/ MF: def: $b_k \equiv c_{-k} c_k$, $b^\dagger_k = c^\dagger_k c^\dagger_{-k}$

$$b^\dagger_k b_{k'} \to b_k = \langle b_k\rangle + \delta b_k = F_k + \delta b_k$$
$$b^\dagger_k = \langle b^\dagger_k\rangle + \delta b^\dagger_k = F^\dagger_k + \delta b^\dagger_k$$

$$\approx F^*_k F_k + F^\dagger_k \delta b_k + F_k \delta b^\dagger_k = \cancel{F^\dagger_k F_{k'}} - \cancel{F^*_k F_{k'}} + F^\dagger_{k'} b_k + F_k b^\dagger_{k'} - F^\dagger_{k'} F_k$$

$$H^{MF} = \sum_k \xi_k (c^\dagger_k c_k + c^\dagger_{-k} c_{-k}) - \sum_k\left[\Big(\underbrace{\sum_{k'} \bar V \langle c^\dagger_{-k'} c^\dagger_{k'}\rangle}_{\Delta^*}\Big) c_k c_{-k} + \Big(\sum_{k'} \bar V \langle c_{k'} c_{-k'}\rangle\Big) c^\dagger_{-k} c^\dagger_k\right] + \bar V \underbrace{\sum_{kk'} \langle\ \rangle\langle\ \rangle}_{\substack{\| \\ \Delta^2/\bar V}} =$$

$cc^\dagger + c^\dagger c = 1$ $\quad \overset{\|}{-c_k c^\dagger_k + 1}$

def: $\Delta_k = \sum_{k'} V_{kk'}\langle c_{k'} c_{-k'}\rangle \simeq \Delta$ za $V_{kk'} \simeq \bar V$

$$= \sum_k (c_k \;\; c^\dagger_{-k})\begin{pmatrix} -\xi_k & -\Delta^*_k \\ -\Delta_k & \xi_k\end{pmatrix}\begin{pmatrix} c^\dagger_k \\ c_{-k}\end{pmatrix} + \left(\frac{\Delta^2}{\bar V} + \sum_k \xi_k\right) = \quad (\text{drugače dobiš } c_k c^\dagger_k)$$

$$= \sum_k (c^\dagger_k \;\; c_{-k})\begin{pmatrix} \xi_k & -\Delta_k \\ -\Delta^*_k & -\xi_k\end{pmatrix}\begin{pmatrix} c_k \\ c^\dagger_{-k}\end{pmatrix} + \left(\frac{|\Delta|^2}{\bar V} + \sum_k \xi_k\right)$$

(?) minus?

Bogoljubov:

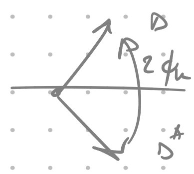
$$= \sum_k (\alpha^\dagger_k \;\; \alpha_{-k})\underbrace{\begin{pmatrix} u & -v \\ v & u\end{pmatrix}\begin{pmatrix} \xi & -\Delta \\ -\Delta^* & -\xi\end{pmatrix}\begin{pmatrix} u & v \\ -v & u\end{pmatrix}}\begin{pmatrix} \alpha_k \\ \alpha^\dagger_{-k}\end{pmatrix}$$

$$\begin{pmatrix} u & -v \\ v & u\end{pmatrix}\begin{pmatrix} \xi u - \Delta v & \xi v + \Delta u \\ \Delta^* u + \xi v & \Delta^* v - \xi u\end{pmatrix} = \begin{pmatrix} \xi u^2 + uv(\Delta^* - \Delta) - \xi v^2 & 2\xi uv + \Delta u^2 - \Delta^* v^2 \\ - \| - \;\; * & \xi v^2 - \xi u^2 + uv(\Delta + \Delta^*)\end{pmatrix}$$

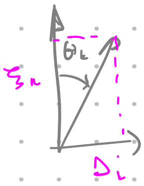
$$u^2 + v^2 = 1 \to \quad u = \cos\frac{\theta_k}{2} \qquad v = \sin\frac{\theta_k}{2} e^{i\phi_k}$$

$$\Delta^* e^{i2\phi_k} = \Delta$$

$$\Rightarrow \quad \xi_k\, 2\sin\left(\frac{\theta_k}{2}\right)\cos\left(\frac{\theta_k}{2}\right) = +|\Delta|\left(\cos\left(\frac{\theta_k}{2}\right)^2 - \sin\left(\frac{\theta_k}{2}\right)^2\right)$$



$\sin(\theta_k)\,\xi_k = |\Delta|\cos\theta_k \Rightarrow$ $\boxed{\tan\theta_k = \frac{|\Delta_k|}{\xi_k}}$ in $\boxed{u_k = \cos\frac{\theta_k}{2}, \quad v_k = \sin\frac{\theta_k}{2}}$



$\rightarrow \cos\theta_k = \frac{\xi}{\sqrt{\Delta^2+\xi^2}}, \quad \sin\theta_k = \frac{\Delta}{\sqrt{\xi^2+\Delta^2}}$ ⊛

$u_k = \cos\frac{\theta_k}{2} = \sqrt{\frac{1+\cos\theta_k}{2}} \rightarrow$ $\boxed{u_k^2 = \frac{E_k+\xi_k}{2E_k}, \quad v_k^2 = \frac{E_k-\xi_k}{2E_k}}$

$$H = \sum_k E_k \alpha_k^\dagger \alpha_k + (\xi_k - E_k) - \frac{\Delta^2}{V}$$

$+$ GS: $\alpha_k|BCS\rangle = 0 \Rightarrow$ $\boxed{|BCS\rangle = \prod_k \left(u_k + v_k c_k^\dagger c_{-k}^\dagger\right)|0\rangle}$

(2) Pauli | Anderson:

p-tolne: $\Delta_k = \frac{V}{2}\sum \sin\theta_k = V\sum\langle c_k c_k\rangle = V\sum\frac{\langle X_k\rangle}{2} = V\sum\langle S_x\rangle$

Self consistency: $\langle\alpha_k^\dagger\alpha_k\rangle = f_k = \frac{1}{e^{\beta E_k}+1} = \frac12 - \frac12\,\text{th}\left(\frac12\beta E_k\right)$

$\Rightarrow \langle c_k^\dagger c_k\rangle = \frac12 - \frac{\xi_k}{2E_k}\text{th}\left(\frac12\beta E_k\right)$

$\langle c_{-k\sigma} c_{k\sigma}\rangle = -\frac{\sigma\Delta_k}{2E_k}\,\text{th}\left(\frac12\beta E_k\right)$

$T=0$ ⊛ $|\xi_k| < \omega_D$

by def: $\Delta_k = \sum_{k'} V_{kk'}\langle c_{k'}c_{k'}\rangle \simeq \frac{V}{2}\sum_{k'}\sin\theta_{k'} = \frac{V}{2}\sum_{k'}{}' \frac{\Delta_{k'}}{\sqrt{\Delta_k^2+\xi_k^2}}$

per spin

$\rightarrow$ $\boxed{\frac{1}{V} = \frac12\int_{-\omega_D}^{\omega_D}\frac{\frac12 g(\xi)}{\sqrt{\Delta^2+\xi^2}}\,d\xi}$ $\rightarrow$ $\boxed{\Delta \simeq 2\omega_D\, e^{-\frac{2}{g(\epsilon_F)V}}}$

by def: $\langle c_{-k}c_k\rangle = \left\langle\left(u_k\alpha_{-k} - v_k\alpha_k^\dagger\right)\left(u_k\alpha_k + v_k\alpha_{-k}^\dagger\right)\right\rangle =$

$= u_k v_k\left(1 - \langle\alpha_k^\dagger\alpha_k\rangle - \langle\alpha_{-k}^\dagger\alpha_{-k}\rangle\right) = \frac12\sin\theta_k\cdot\text{th}\left(\frac{\beta E_k}{2}\right)$

$\cos\frac{\theta_k}{2}\sin\frac{\theta_k}{2} = \frac12\sin\theta_k$

$1 - f_k - f_{-k} = 1 - \frac{2}{e^{\beta E_k}+1}$

- od tuje self. cons.: (1) Δ arbitraren, ne vemo kaj je $\langle cc\rangle$, odvisno od vpliva C.P.
  (2) Ampak Δ je vključena v $H^{MF}(\Delta)$!

loop: $\Delta \rightarrow H^{MF}(\Delta) \rightarrow |\text{stanje}(\Delta)\rangle \rightarrow \langle c_k c_k\rangle \rightarrow \Delta'$ repeat.

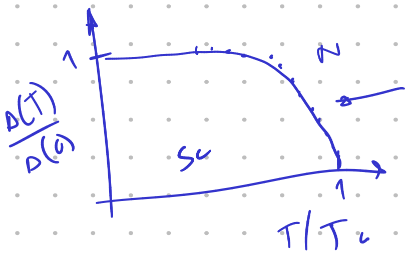
→ vsak C.P. čuti kolektivni vpliv ostalih parov in si ustvari svoj pričakovan $\langle cc\rangle$, ker prispeva k ostalim, ki spet spremenijo stanje, kar spremeni kako stanje C.P.

$T>0$: vemo $\langle c_k c_{-k}\rangle = \frac{1}{2}\sin\theta_k \, th\left(\frac{\beta E_k}{2}\right) = \frac{\Delta_k}{2E_k} th\left(\frac{\beta E_k}{2}\right)$

$$\Rightarrow \quad \Delta \simeq \frac{\bar V}{2}\sum_k \sin\theta_k \, th\left(\frac{\beta E_k}{2}\right) = \frac{\bar V \Delta}{2}{\sum_k}' \frac{th\left(\frac{\beta}{2}\sqrt{\Delta^2+\xi_k^2}\right)}{\sqrt{\Delta^2+\xi_k^2}} \Rightarrow$$

$$\Rightarrow \quad \boxed{\frac{1}{\bar V} \simeq \frac{1}{2}\int_{-\omega_D}^{\omega_D} d\xi \, \frac{1}{2} g(\epsilon_F) \frac{th\left(\frac{\beta}{2}\sqrt{\Delta^2+\xi^2}\right)}{\sqrt{\Delta^2+\xi^2}}} \to \Delta(T) = \dots$$

per spin!



$\Delta(T) = 0$ vemo.

numerično: ① $$\Delta'_{n+1} = \frac{g(\epsilon_F)\bar V}{2}\cdot \Delta_n \cdot \sum_{\xi=0}^{\omega_D} \overbrace{\left(\frac{\omega_D}{N}\right)}^{d\xi} \frac{th\left(\frac{\beta}{2}\sqrt{\Delta_n^2+\xi^2}\right)}{\sqrt{\Delta_n^2+\xi^2}}$$

② $\Delta_{n+1} = (1-a)\Delta_n + a\Delta'_{n+1}$ (stabilnost)!

③ če $|\Delta_{n+1}-\Delta_n| < \varepsilon$ stop..

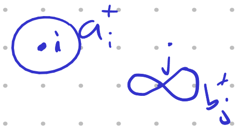
efektivno imamo iteracijo: $\Delta_{n+1} = A(T)\Delta_n$; če $A(T)>1 \Rightarrow \Delta>0$

če $A(T)<1 \Rightarrow \Delta=0$

pri $T=T_c \to A(T)=1$

Konvergenca počasna okoli $A=1$!

2) EXCITON INSULATOR:

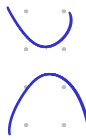


→ Coulomb, interband

ij mesti, a, b orbitali,

$$\rightarrow H = H_0 + H_V = \sum_{\substack{ij \\ \sigma = a,b}} \left(\varepsilon^{(\sigma)} \delta_{ij} + t_{ij}^{(\sigma)}\right) a_i^+ a_j + V \sum_j a_j^+ a_j b_j^+ b_j =$$

$$= \sum_k \varepsilon_a(k)\, a_k^+ a_k + \varepsilon_b(k)\, b_k^+ b_k + \underbrace{V \sum_j a_j^+ a_j b_j^+ b_j}$$



$\varepsilon_a(k)$, $\varepsilon_b(k)$ } non-interacting

lokalna interakcija, on site, lahko spremenimo!

MF za H_V:

$$a_j^+ a_j b_j^+ b_j \simeq \underbrace{\langle a_j^+ a_j \rangle}_{n_a} b_j^+ b_j + \underbrace{\langle b_j^+ b_j \rangle}_{n_b} a_j^+ a_j - n_a n_b \quad + \; \leftarrow \text{Hartree}$$

$$- \underbrace{\langle a_j^+ b_j \rangle}_{\phi} b_j^+ a_j - \underbrace{\langle (a_j^+ b_j)^+ \rangle}_{\phi^*} a_j^+ b_j + \langle a_j^+ b_j \rangle \langle b_j^+ a_j \rangle \leftarrow \text{Fock/exchange}$$

$a^+ a b^+ b = -a^+ b^+ a b = (-1)^3 a^+ b\, b^+ a$

od fermionij

elektroni/density $\hat{n}_b$ čuti average density v drugem bandu.

$$H_V^{MF} = V n_b \sum_j a_j^+ a_j + V n_a \sum_j b_j^+ b_j - V\phi \sum_j b_j^+ a_j - V\phi^* \sum_j a_j^+ b_j + NV\left(|\phi|^2 - n_a n_b\right)$$

→ FT: $a_j^+ = \frac{1}{\sqrt{N}} \sum_k e^{ikR_j} a_k^+$ etc.

→ $n_a = \langle a_i^+ a_i \rangle = \frac{1}{N} \sum_{kk'} e^{i(k-k')R_i} \langle a_k^+ a_{k'} \rangle = \frac{1}{N} \sum_k \langle a_k^+ a_k \rangle$

→ $\phi = \langle a_j^+ b_j \rangle = \frac{1}{N} \sum_k \langle a_k^+ b_k \rangle$ → $\Delta \simeq -\frac{V}{N} \sum_k \langle a_k^+ b_k \rangle$, $\Delta^* = -\frac{V}{N} \sum_k \langle b_k^+ a_k \rangle$

$$\parallel$$

$$-\frac{1}{N} \sum_k V(k-k') \langle a_k^+ b_k \rangle.$$

$$V(\vec{q}) = \int e^{i\vec{q}\vec{r}} V(\vec{r})\, d\vec{r} \qquad N \frac{NV}{N} = N^2 \bar{V}$$

na koncu:

$$H^{MF} = \sum_k \left(\left[\varepsilon_a(k) + V n_b\right] a_k^+ a_k + \left[\varepsilon_b(k) + V n_a\right] b_k^+ b_k + \Delta^* a_k^+ b_k + \Delta b_k^+ a_k \right) + \left(\frac{|\Delta|^2}{\bar{V}} - \bar{V} N^2 n_a n_b \right)$$

$\left(\bar{V} = \frac{V}{N}\right)$ enako kot pri BCS.

$$H^{MF} = \sum_k \left(a_k^+ \; b_k^+\right) \begin{pmatrix} \overbrace{\varepsilon_a + V n_b}^{\bar{\varepsilon}_a} & \Delta^* \\ \Delta & \underbrace{\varepsilon_b + V n_a}_{\bar{\varepsilon}_b} \end{pmatrix} \begin{pmatrix} a_k \\ b_k \end{pmatrix} + \text{const.}$$

bands: $E_\pm = \frac{\bar{\varepsilon}_a + \bar{\varepsilon}_b}{2} \pm \sqrt{\left(\frac{\bar{\varepsilon}_a - \bar{\varepsilon}_b}{2}\right)^2 + \Delta^2}$

self consistency ET :

$$\begin{pmatrix}a_k\\ b_k\end{pmatrix} = \begin{pmatrix}u & v\\ -v & u\end{pmatrix}\begin{pmatrix}\alpha_k\\ \beta_k\end{pmatrix}$$ } samo mix med dvema bandoma

$$\begin{pmatrix}a_k^\dagger & b_k^\dagger\end{pmatrix} = \begin{pmatrix}\alpha_k^\dagger & \beta_k^\dagger\end{pmatrix}\begin{pmatrix}u & -v\\ v & u\end{pmatrix}$$

$$\to \mathcal{H}_k = \begin{pmatrix}u & -v\\ v & u\end{pmatrix}\begin{pmatrix}\bar\varepsilon_a & \Delta^*\\ \Delta & \bar\varepsilon_b\end{pmatrix}\begin{pmatrix}u & v\\ -v & u\end{pmatrix} =$$

$$= \begin{pmatrix}u & -v\\ v & u\end{pmatrix}\begin{pmatrix}\bar\varepsilon_a u - \Delta^* v & \bar\varepsilon_a v + \Delta^* u\\ \Delta u - v\bar\varepsilon_b & \Delta v + \bar\varepsilon_b u\end{pmatrix} = \begin{pmatrix}\bar\varepsilon_a u^2 - \Delta^* uv + v^2\bar\varepsilon_b - \Delta uv & (\bar\varepsilon_a-\bar\varepsilon_b)uv + \Delta^* u^2 - \Delta v^2\\ -\|-^* & \bar\varepsilon_a v^2 + \bar\varepsilon_b u^2 + \Delta^* uv + \Delta uv\end{pmatrix}$$

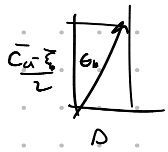
$$1 = \{a_k, a_k^\dagger\} = \{u\alpha + v\beta,\ u^*\alpha^\dagger + v^*\beta^\dagger\} = u^2 + v^2 \to u_k^2 + v_k^2 = 1 \Rightarrow u_k = \cos\frac{\theta_k}{2}$$

$$\Rightarrow v_k = \sin\frac{\theta_k}{2}e^{i\varphi_k}$$

$$\Rightarrow \frac{(\bar\varepsilon_a-\bar\varepsilon_b)}{2}\cdot\sin\theta_k + \Delta\cos\theta_k = 0 \to \tan\theta_k = \frac{-2\Delta}{\bar\varepsilon_a-\bar\varepsilon_b}$$

$\hookrightarrow u_k v_k = -\frac{\sin\theta_k}{2}$

$\Delta^* = \Delta e^{i\varphi_k}$



$$\Rightarrow \langle a_k^\dagger b_k\rangle = \langle (u_k\alpha_k^\dagger + v_k\beta_k^\dagger)(-v_k\alpha_k + u_k\beta_k)\rangle = u_k v_k\left(\langle\beta_k^\dagger\beta_k\rangle - \langle\alpha_k^\dagger\alpha_k\rangle\right) =$$

$$= -\frac{1}{2}\sin\theta_k\left(-f_\alpha + f_\beta\right)$$

$$\Delta = +V\sum_k\langle a_k^\dagger b_k\rangle = -\frac{V}{2}\sum_k\frac{\Delta}{\sqrt{\underbrace{\left(\frac{\varepsilon_a-\varepsilon_b}{2}\right)^2}_{\xi_k^2}+\Delta^2}}\cdot(f_\alpha - f_\beta)$$

$f_\alpha = f(E_+)$

$$\to \frac{1}{V} = -\frac{1}{2}\sum_k\frac{1}{\sqrt{\xi_k^2+\Delta^2}}\cdot\left(f(E_+) - f(E_-)\right)$$

ker je $H\begin{pmatrix}u_k\\ -v_k\end{pmatrix} = E_+\begin{pmatrix}u_k\\ -v_k\end{pmatrix} \to H = \sum_k E_+\alpha_k^\dagger\alpha_k + E_-\beta_k^\dagger\beta_k +$ skalar

$\langle\alpha_k^\dagger\alpha_k\rangle = f(E_+)$!

če $\bar\varepsilon_a = -\bar\varepsilon_b$ aka: $n_a = n_b$

potem $E_\pm = 0 \pm \sqrt{\xi_k^2+\Delta^2} \to$ se nanaša na BCS:

$$f(E_-) - f(E_+) = f(-E) - f(E) = \frac{1}{e^{\beta(-E-\mu)}+1} - \frac{1}{e^{\beta(E-\mu)}+1} = 1 - 2f(E) = \text{th}\left(\frac{\beta E}{2}\right)$$

$E_- = -\sqrt{\xi_k^2+\Delta^2}$

$E_+ = +\sqrt{\xi_k^2+\Delta^2}$

$\hookrightarrow f(-E) = 1 - f(E)$

numerika za EI gap:

- $E_\pm = E_\pm(n_a, n_b, \Delta) = \dfrac{\varepsilon_a(k) + \varepsilon_b(k) + V(n_a + n_b)}{2} \pm \left(\left[\dfrac{\varepsilon_a(k) + \varepsilon_b(k) + V(n_a+n_b)}{2} \right]^2 + |\Delta|^2 \right)^{1/2}$

- $f(E_\pm, \mu) = \dfrac{1}{\exp(\beta(E_\pm - \mu)) + 1}$ ← $n = n_a + n_b = \text{const.} \Rightarrow \mu \neq \text{const.}$

$$N = \int_{-\infty}^{\infty} d\varepsilon_a \, f(E_+) g(\varepsilon_a) + \int_{-\infty}^{\infty} d\varepsilon_b \, f(E_-) g(\varepsilon_b) = \text{const.} \Rightarrow \mu^{(\ell)} = \dots$$

- gap:
$$\Delta^{(\ell+1)} = \frac{\bar{V}}{2} \Delta^{(\ell)} \sum_k \frac{1}{\left(\Delta^{(\ell)2} + \left[\dfrac{\varepsilon_a - \varepsilon_b + V(n_a^{(\ell)} - n_b^{(\ell)})}{2} \right]^2 \right)} \left[f(E_-^{(\ell)}) - f(E_+^{(\ell)}) \right].$$

init $(\Delta, n_a, n_b) \rightarrow (\bar{\varepsilon}_a, \bar{\varepsilon}_b) \rightarrow (E_+, E_-) \rightarrow \mu \xrightarrow{\text{gap eq.}} (\Delta_{nov}, n_a^{nov}, n_b^{nov})$

dolocimo $n = \text{const.}$

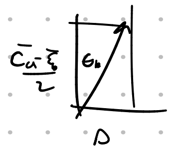
n_a, n_b updated with:

$$n_a = \frac{1}{N} \sum_k \left(u_k^2 f(E_+) + v_k^2 f(E_-) \right)$$
$$n_b = \frac{1}{N} \sum_k \left(v_k^2 f(E_+) + u_k^2 f(E_-) \right)$$

$u_k, v_k = u_k, v_k(\Delta, n_a, n_b)$

$\tan\theta_k = \dfrac{-2\Delta}{\bar{\varepsilon}_a - \bar{\varepsilon}_b}$

$\hookrightarrow u_k v_k = \dfrac{-\sin\theta_k}{2}$



$u_k v_k = -\frac{1}{2} \sin\theta_k$

$\tan\theta_k = -\dfrac{\Delta}{\xi_k}$

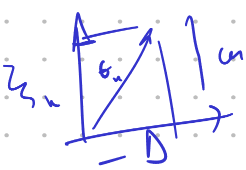
def
$$\boxed{\begin{aligned} \xi_k &= \frac{\bar{\varepsilon}_a - \bar{\varepsilon}_b}{2} \\ \eta_k &= \frac{\bar{\varepsilon}_a + \bar{\varepsilon}_b}{2} \end{aligned}}$$

$$\boxed{\begin{aligned} \eta_k &= \eta_k(n_a, n_b) \\ \xi_k &= \xi_k(n_a, n_b) \end{aligned}}$$

$\rightarrow u_k^2 + v_k^2 = 1 \qquad -\dfrac{\sin\theta_k}{2} = c\cos\theta_k \dfrac{\Delta}{2\xi_k} = u_k v_k$

$$u_k^2 = \cos^2\frac{\theta_k}{2} = \frac{1}{2}(1 + \cos\theta_k) = \frac{1}{2}\left(1 + \frac{\xi_k}{\sqrt{\Delta^2 + \eta_k^2}}\right) = \frac{1}{2}\left(1 + \frac{\xi_k}{\sqrt{\Delta^2 + \xi_k^2}}\right) = u_k^2$$

$$v_k^2 = \frac{1}{2}\left(1 - \frac{\xi_k}{\sqrt{\Delta^2 + \xi_k^2}}\right)$$



$\cos\theta_k = \dfrac{\xi_k}{\sqrt{\Delta^2 + \xi_k^2}}$

$$n_a = \frac{1}{N}\sum_k \langle a_k^+ a_k \rangle = \langle (u_k \alpha_k^+ + \beta_k^+ v_k)(u_k \alpha + v\beta) \rangle = \frac{1}{N}\sum_k \left[u_k^2 f(E_+) + v_k^2 f(E_-) \right] = n_a$$

$$\langle \alpha^+ \alpha \rangle = f(E_+)$$

$$\frac{1}{N}\sum_k \left[v_k^2 f(E_+) + u_k^2 f(E_-) \right] = n_b$$

$$n_a + n_b = \frac{1}{N}\sum_k \left[f(E_+) + f(E_-) \right]$$

again:

def: ① $\varepsilon_a(k)$, $\varepsilon_b(k)$ (rdeče, zeleno)

② (init n_a, n_b) $\bar{\varepsilon}_a = \varepsilon_a(k) + V n_b$, $\bar{\varepsilon}_b = \varepsilon_b(k) + V n_a$ $\Rightarrow$ $\eta_k = \frac{\bar{\varepsilon}_a + \bar{\varepsilon}_b}{2} - \mu$?, $\xi_k = \frac{\bar{\varepsilon}_a - \bar{\varepsilon}_b}{2}$

③ $E_\pm = \eta_k \pm \sqrt{\xi_k^2 + \Delta^2}$ (init Δ)

④ za init. normir.: $\text{const.} = n_a + n_b = \frac{1}{N}\sum_k \left(f(E_+) + f(E_-) \right) \to$ določi μ.

1' half f.

⑤ gap eq:

$$\Delta' = \frac{V}{2}\Delta \cdot \sum_k \frac{f(E_-) - f(E_+)}{\sqrt{\xi_k^2 + \Delta^2}} \to \text{update } \Delta \to \Delta' \quad (E_\pm \to E_\pm')$$

⑥ update n_a, n_b: • ustrezno $u_k'^2 = \left(1 + \frac{\xi_k}{\sqrt{\xi_k^2 + \Delta'^2}} \right)$, $v_k'^2 = (-\text{''} \text{ z minusom})$

• $n_a' = \frac{1}{N}\sum_k \left[u_k'^2 f(E_+') + v_k'^2 f(E_-') \right]$, podobno n_b'

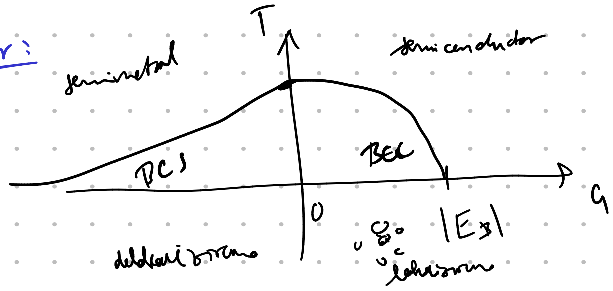
$n_a \to n_a'$, $n_b \to n_b'$ (tudi $\eta_k \to \eta_k'$, $\xi_k \to \xi_k'$)

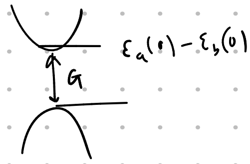
⑦ • update $\mu' \Leftarrow n_a' + n_b' = \text{const.}$ $\mu \to \mu'$

⑧ nazaj v gap eq. $\Delta'' = \ldots$

1dim tight binding toy model: N sites, PBC.

BCS – BEC crossover:





če imamo zamaknjene pasove → exciton z com momentum $\vec{q}$ $\Rightarrow$ $\Delta_{\vec{q}} \propto \langle a^{\dagger}_{\vec{k}+\vec{q}}\, b_{\vec{k}} \rangle$.